

Hyper Suprime-Cam: Filters

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Abstract

We describe five broad-band filters and a narrow-band one for the Hyper Suprime-Cam (HSC) at the prime focus of the Subaru Telescope. Since the HSC is a practical successor to the Suprime-Cam (SC), our broad-band filter set follows the SC's SDSS g' , r' , i' , z' filters, and Y filter, and the $H\alpha$ filter is adopted as our first narrow-band filter. Filters for the HSC are quite large; therefore, they are constructed only by interference coatings. Based on a draft specification guided by that of SC's filters, we made prototype filters which were evaluated at a laboratory, and eventually used them in the actual HSC commissioning observation. Through this process, we learned that transmission measurement with a fine spatial sampling was crucial for reliable understanding of the filter characteristics, and this motivated us to develop an efficient measurement system. Since this measurement system is placed in the summit facility and functions semi-automatically, long-term monitoring of the performance of HSC filters will be achieved. We present in this paper the specifications that the HSC filter should meet, problems and solutions in filter implementation, issues found through the commissioning observing run, the new filter measurement system at the summit facility, and transmission curves and tables of the filters.

Key words: instrumentation: miscellaneous — techniques: image processing — techniques: miscellaneous

1 Introduction

Hyper Suprime-Cam (HSC; Miyazaki et al. 2018) is an 800-megapixel-wide field camera built for the prime focus of the Subaru Telescope. The HSC has a 1.5° -diameter field of view which is unique among 8–10 m large telescopes and therefore it is powerful in wide field survey projects. Since the HSC is a practical successor to the Suprime-Cam (SC; Miyazaki et al. 2002), the broad-band filter set of HSC should follow the filter system of SC.

For general applications, the SC has nine broad-band filters (Johnson–Cousins B , V , R_C , I_C , SDSS g' , r' , i' , z' plus Y). It also has a rest-frame $H\alpha$ filter centered at 656 nm (NA656) and a set of narrow-band filters (NB711, NB816, NB921) prepared for surveys of $\text{Ly}\alpha$ emitters.¹ Since most modern surveys employ the Sloan Digital Sky Survey (SDSS) photometric system, we also adopted the SDSS system for the broad-band filter set of HSC.

¹ (<http://www.naoj.org/Observing/Instruments/SCam/sensitivity.html>).

We selected silica glass with a diameter of 600 mm and a thickness of 15 mm as the substrate of the HSC filter. We also adopted *B270* glass because its performance is equivalent to silica but it is cheaper than silica (see subsubsection 2.2.2). The *B270* filter has the same diameter as the silica but has a thickness of 13.7 mm, chosen to compensate for the difference of refraction index between silica and *B270*. The filter glass is mounted in a special filter frame which is loaded in the filter exchanger of HSC. This filter frame has dimensions of 710 mm $\times$ 660 mm $\times$ 42 mm.

The HSC Subaru Strategic Program (hereafter HSC-SSP) is the largest survey program ever carried out by the Subaru Telescope. Three hundred nights over a span of five years are allocated for the HSC-SSP. In order to guarantee the stability of the survey observations, it is crucial to monitor the wavelength response function and the absolute throughput of the instrument periodically. As our measurement system for the filter acceptance test is located at the Mitaka campus of the National Astronomical Observatory of Japan (NAOJ), we decided to build a new filter measurement system in the summit facilities of the Subaru Telescope as part of the long-term sensitivity monitoring program.

Some wavelength shifts were observed in the filters of SDSS (Doi et al. 2010). Most of these shifts were caused by changes in environmental conditions. Since the filters of SDSS are located in vacuum chamber, most of them have shown sufficiently stable characteristics. However, the response function of the SDSS-*u* filter has shown significant change in the long term. The timescale of this variability is about three years, so it is desirable to monitor the response function of the HSC filters periodically, every two years at least.

Because the HSC filters are located in a very fast incoming beam of $F/2.25$, the light penetrating the filter has various incident angles. Since all HSC filters are produced by using interference coating technology, the property of transmittance is affected by the incident angle of light. The $F/2.25$ beam has a 12.5° opening angle, so the rays in the beam have 0° to 12.5° incident angles and the corresponding wavelength shift is up to several nanometers. Therefore our measurement system should mimic the fast beam of HSC.

Finally, it should be mentioned that the measurement system will be operated at the summit facility, whose elevation is about 4200 m. The oxygen partial pressure in the summit facility is about 60% of that at sea level. Thus the measurement system should work as automatically as possible to avoid human error in operating the system.

The following sections describe the HSC filters and the transmission measurement system. The filter specification and implementation are described in section 2, and

details of the filter measurement system are demonstrated in section 3. A flat-field error in the commissioning data and its cause and countermeasure are explained in section 4, and section 5 shows brief filter properties.

2 Filter production

2.1 Specification

These filters consist of multi-layer interference coatings deposited on plane-parallel synthetic silica substrates or equivalents. We present detailed specifications for the broad-band filters in this section.

2.1.1 Physical properties

Since the HSC has about a three times larger focal plane diameter than SC, it is inevitable that the requirement for the physical size of the filters becomes large in the same proportion. The clear aperture size of the SC filters is 192 mm $\times$ 158 mm, so we adopt 600 mm as the diameter of the substrates. The clear aperture of the filter is 574 mm. Also, the back-focus of the wide-field corrector (WFC) is only about 180 mm, and the thickness of the filter frame is very constrained because the focal plane shutter, the filter, and the dewar window are also placed in this limited space. We tabulate the requirements for these physical properties as specifications in table 1.

2.1.2 Optical properties

We define a sub-aperture (SA) as the cross section of the converging beam bisected by the filter surface, and the diameter of the SA is about 26.5 mm. This size of SA is equivalent to the beam size of the light from a point source in the field of view of HSC. The WFC has a non-telecentricity of about 7.7° at a field angle of 0.75° ; however, the inclination of the outermost ray in the effective beam is only 14.6° because of optical vignetting. Since the $F/2.25$ beam is a circular cone that has 12.5° half-opening angle, the non-telecentricity of the WFC has a small impact on the location dependency of the SA.

The transmitted wavefront error of a filter should be determined not to affect the image quality in any SA in the filter surface. Since there is about 200 nm RMS wavefront error at 760 nm in the optical design of the WFC, we adopt a value of 100 nm RMS as a representative threshold of the transmitted wavefront error of the filter substrate. The WFC optics are made to achieve image quality in a $0.3''$ diameter of 80% encircled energy (D80). The 100 nm wavefront error of the substrate deteriorates the D80 to $0.33''$ when it is added to that of the WFC in quadrature. This is adequately smaller than the typical seeing size at Mauna Kea, $0.6''$.

Table 1. Requirements for physical, optical, and spectral properties for the filter set.*

Classification	Item	Specifications	Goal	Suprime-Cam
Physical	Substrate	Synthetic silica		BK7 & color glass
	Diameter	600 mm		205 × 170 mm
	Weight	<25 kg (w/frame)		
	Clear aperture (CA)	574 mm		192 × 158 mm
	Thickness	<42 mm		20 mm (w/holder)
Optical	Thickness error	<0.5 mm		<0.2 mm
	Diameter of sub-aperture (SA)	25 mm		30 mm
	Wavefront error in SA	100 nm RMS	30 nm RMS	
Spectral	Peak transmittance	>90%	>95%	>95%
	Out-of-band leak	<0.1% 300–350 nm	<0.1% 300–350 nm	<0.1%
		<0.01% 350–1000 nm	<0.01% 350–1000 nm	
	Cut-on/-off wavelength error	<2% of central wavelength	<0.5%	~0.6%–1.0%
	Uniformity (wavelength)	<2% of central wavelength	<0.5%	~0.4%
	Uniformity (transmission)	<2%	<0.5%	~1%
	In-band transmission variation	<5%	<2%	~5%

*We set a “Goal” that might not currently be achieved but is eventually expected after the improvements of the coating facilities. In-band variation of transmission is not applied for the H α filter.

Finally, we should mention that the thickness of the filter is taken into account in the optimization of the optical design of the WFC, so the effect of the thickness error of the filter on the image quality must be considered. By using ray tracing, we confirm that image deterioration owing to a 0.5 mm thickness error can be acceptable. We tabulate the requirements for the optical properties as specifications in table 1.

2.1.3 Spectral properties

We chose to equip the HSC instrument with only SDSS (and not Johnson) filters because much of the astronomical community has tended to use this system for photometric observations. Typical values of the characteristics for the SDSS filters of SC are shown in the last column of table 1. Since the WFC optics have relatively small transmission at wavelengths below 350 nm (Miyazaki et al. 2018) and the detectors have very low sensitivity at wavelengths above 1050 nm (Kamata et al. 2014), we chose 300–1100 nm as the wavelength range in which the out-of-band light leak is minimized. In the wavelength range of 300–350 nm we relax the specification for leakage by a factor of 10, because the transmission of the WFC is almost zero in this wavelength range since the anti-reflection coating on the optical elements is not optimized to the wavelength range of 300–350 nm. Therefore, the filter has only to block environmental ultraviolet light incoming through the filter insertion slot of HSC, which is opened to the outside during the period of observation. We tabulate the requirements for the spectral properties as specifications in table 1.

Table 2. Expected wavelength coverage of the HSC broad-band filter set along with the data for the SC filter set.

HSC filter	Wavelength [nm]	SC filter	Wavelength [nm]
<i>g</i>	400–550	<i>g'</i>	400–539
<i>r</i>	550–695	<i>r'</i>	552–695
<i>i</i>	695–845	<i>i'</i>	689–845
<i>z</i>	845–930	<i>z'</i>	849–989
<i>y</i>	930–1070	<i>Y</i>	969–1068

*We use the SC filter data at the Subaru Telescope web site.²

2.1.4 Summary of specifications

The measured values of the SC filters are also shown for reference. We tried to follow the specifications of the SC filters, but in some cases we were forced to relax the requirements because no coating vendor could guarantee the quality on such a large substrate in the initial phase. We plan to replace the filter to meet the goal specification shown in the table as the manufacturing capability of the vendors is improved.

2.1.5 Expected transmission curves

We present the expected wavelength coverage of the HSC broad-band filter set in table 2 along with those of the SC filters. The full-width at half maximum of the filter response function is used as the wavelength coverage. For illustration, we present the expected transmission curves of the broad-band filter set in figure 1.

² (<http://www.naoj.org/Observing/Instruments/SCam/sensitivity.html>).

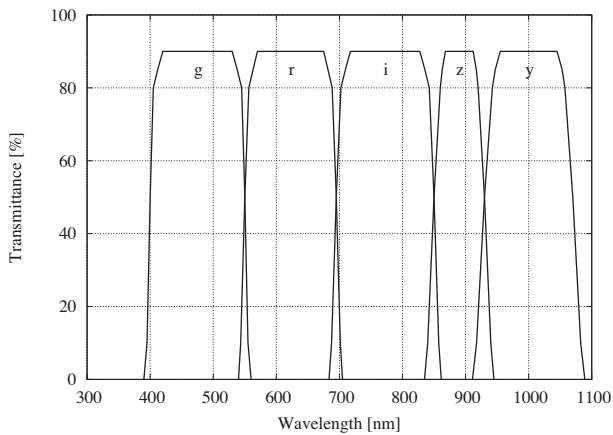


Fig. 1. Expected transmission curves for the HSC filter set.

The wavelength coverage of the SC filters follows that of SDSS. We want to follow the SC filters, but the wavelength coverage should not have gaps or overlaps at their boundary. Therefore, we adopt these values of wavelength coverage for our filter set. The boundary between z and y in HSC is moved to 930 nm, by about 50 nm shorter than SC. This is because the water-vapor absorption band at 950 nm is excluded from the z band and is included in the y band in order to compensate for the poor sensitivity of CCDs at the longer-wavelength edge of the y band. This is a different choice from other standard filter sets, such as SDSS and the Dark Energy Survey.

2.2 Implementation

When we initially contacted optic firms for the drafted specification, no vendor could guarantee uniformity of the performance over such a large substrate or the capability of the out-of-band light leak. Therefore, we decided to ask several vendors for prototypes.

2.2.1 Selection of filter production method

We adopted a standard evaporation deposition for our first filter prototypes. This method could realize a uniform coating on a large substrate; however, there are two problems in using standard evaporation deposition coating. The standard evaporation deposition only realizes porous membranes of limited thickness. A porous coating can easily absorb water vapor in the air and their cut-on/off wavelengths could be easily shifted by a few nanometers. The other problem is that the restriction on the thickness of the coating would force us to require several substrates to be deposited, which is mechanically not favorable.

We therefore had to look for vendors capable of more advanced coating techniques, such as magnetron sputtering (MS), ion-beam-assisted deposition (IAD), and ion plating.

Since these methods can build dense coating layers on a substrate, two surfaces of a substrate are sufficient to satisfy our specifications and the resistance to the environment ought to be better than those of evaporation deposition coatings. In addition to the coating technique, there are several ways to hold and move substrates in the coating chamber. On-axis rotation is commonly used, and planetary rotation (PR) is also used to make more uniform coatings than simple on-axis rotation.

2.2.2 Substrates

The HSC WFC is designed on the premise that a filter substrate is made of silica with a thickness of 15.0 mm. To reduce the production cost of filters we tried to find substitutes for silica. Schott B270[®] superwhite glass has been well known as a substrate for small filters. We found that this material can be used as the substrate for the HSC filter after some measurements of transmitted wavefront error using the Optical shop of the Advanced Technology Center (ATC) at NAOJ. The difference in refractive index between silica and B270 can be compensated for by reducing the thickness to 13.7 mm.

2.2.3 Humidity effect

At the beginning of the prototyping we tested the effect of water vapor in the air on the cut-on/off wavelengths of a filter made by the standard deposition. We exposed a small test piece of filter to a high-humidity environment for 24 hr and measured the transmission curve of the test piece by using a Shimadzu SolidSpec 3700 monochrometer in the ATC. After that, we exposed the test piece to a low-humidity environment for 60 hr and measured the transmission curve again. Because we found 1.6 and 3.5 nm wavelength shifts in the cut-on/off wavelengths, we determined that our filter should be produced by an advanced coating technique, such as sputtering, IAD, or ion plating. We performed a similar test on a small piece of a filter made by IAD, and found that the test piece shows the same values in cut-on and cut-off wavelengths. A summary of the evaluations of environmental effects is given in table 3.

2.2.4 Filter glass holder assembly

A filter frame assembly with the aluminum cover for shipping is shown in figure 2, and a detailed cross-sectional view to show how the filter is held is shown in figure 3. Poly-tetrafluoroethylene (PTFE) pads are disposed between the glass substrate and the filter holder to avoid direct contact between these materials. The urethane pads between the filter retainer and the glass substrate have enough friction under an appropriate preload to fix the glass substrate in a proper position in the filter holder. The filter frame has 48 rollers on its outer edges to guide the filter frame during

Table 3. Humidity effect on the cut-on/off wavelength.*

	T_e [K]	RH [%]	P_{WV} [hPa]	Cut-on [nm]	Cut-off [nm]
Standard, high humidity	293	85	19.9	547.7	693.0
Standard, low humidity	293	10	2.3	546.1	689.5
IAD, high humidity	297	65	19.4	544.9	694.4
IAD, low humidity	297	0	0.0	544.9	694.4

* T_e , RH, and P_{WV} are the environmental temperature, relative humidity, and water vapor partial pressure, respectively.

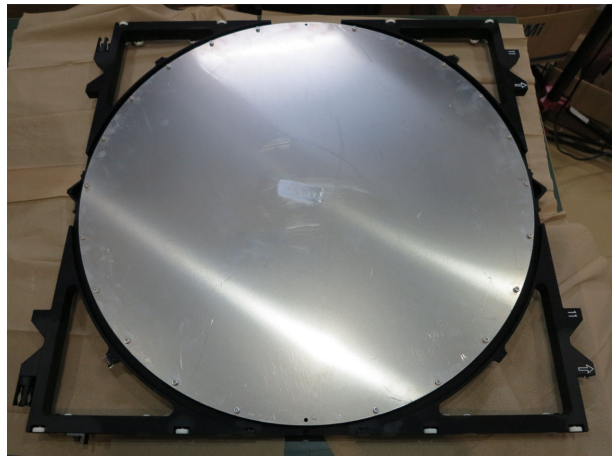


Fig. 2. Filter frame assembly with aluminum cover on top of the filter. (Color online)

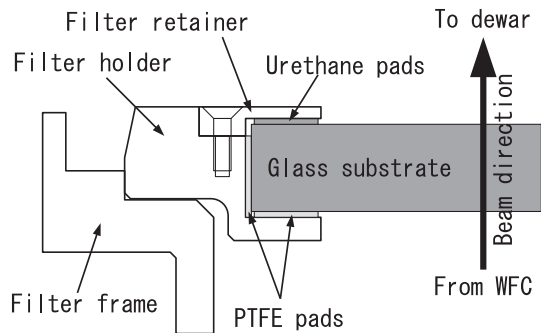


Fig. 3. Close-up cross-sectional drawing of the filter assembly architecture.

a filter exchange. A frame is produced for each filter. Once assembled, the filter and its frame are not disassembled unless we have a particular reason such as a malfunction.

3 New measurement system

3.1 System hardware

The system consists of a broad-band light source with an internal shutter, optics that simulates an F/2.25 beam, a filter rack with a motorized linear stage, an integration sphere, a spectrometer, and a control computer. A schematic view of system components is shown in figure 4, and a photograph of the system is presented in figure 5.

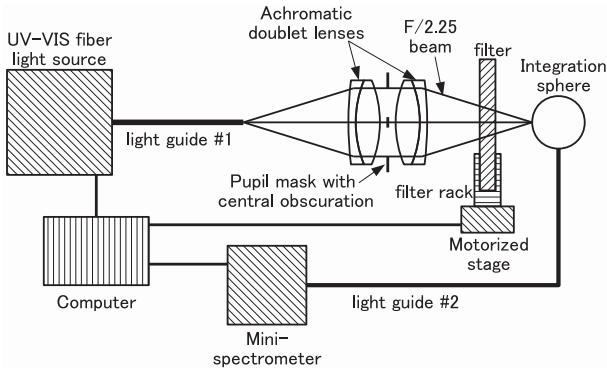


Fig. 4. Schematic view of the system.

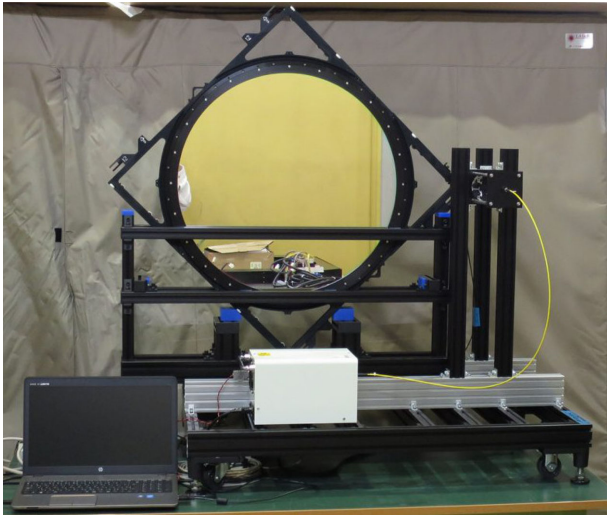


Fig. 5. Photo of the system. A test narrow-band filter with a filter frame is held at the 45° position. Only the blue nylon pads on the filter rack contact the filter frame. (Color online)

A UV-VIS fiber light source (Hamamatsu Photonics L10290³) is used as a broad-band light source. This source has a tungsten-halogen lamp and a deuterium lamp in it, and a shutter is incorporated in front of a light outlet port. A solarization-resistant optical fiber A7969-08AS⁴ is used as a light guide to the beam simulator.

The simulator optics is composed of a pair of identical achromatic doublet lenses that have a focal length

³ (<https://www.hamamatsu.com/jp/en/L10290.html>).

⁴ (https://www.hamamatsu.com/resources/pdf/etd/L10290_TLSZ1010E.pdf).

of 150 mm and a diameter of 80 mm. These lenses are optimized in the visible wavelength range; however, there is no significant degradation of spot size in the near-infrared wavelength region. In the near-ultraviolet wavelength region, the absorption by the flint glass in the doublet is not negligible ($\sim 30\%$ at 10 mm thickness), but it is an acceptable value for our purposes. The continuum level at 400 nm is about 10% of that at the peak (700 nm), and the expected S/N ratio of the continuum is about 100 in a 1000 ms integration time.

The light output of the light guide connected to the light source (light guide #1) is located at the focus of an achromatic lens to make a collimated beam. In the collimated beam, a pupil mask is placed. The diameter of the mask aperture is set to 67 mm to simulate the F/2.25 HSC beam.

After the condenser lens, the filter to be measured is located in the converging beam. The beam size at the surface of the filter is about 26.5 mm. A motorized linear stage that has 500 mm stroke holds the filter rack. As the filter rack can support a filter in eight different directions, a measurement point can be located at eight radial directions on the surface of a filter. The rays that passed through the filter focus on a point, and the entrance port of an integration sphere (IS) is aligned with the focal point of the condenser lens. The IS absorbs the focus change that is caused by the refraction of the plane-parallel filter. Therefore, it is not necessary to adjust the optics in the case of taking continuum data without any filter.

An optical fiber (light guide #2) is installed at the output port of the IS, and the light is fed to a mini-spectrometer (Hamamatsu Photonics C10083CAH⁵). The specifications of the spectral response range, typical wavelength resolution, and wavelength reproducibility of the spectrometer are 320 to 1000 nm, 1 nm, and 0.2 nm, respectively. As the spectrometer uses a linear CCD with 2048 pixels, a full wavelength range spectrum can be achieved by one data readout. The shutter in the light source, the motorized linear stage, and the spectrometer are connected to the control computer for the automation of the overall measurement sequence.

3.2 Measurement sequence

Since the ambient temperature of the system is not controlled, the dark level of the spectrum is not constant. Therefore, we take a dark spectrum by keeping the shutter of the light source closed. It is also necessary to monitor the output level of the light by taking the reference spectrum without filter, although the specification of the light source is fairly

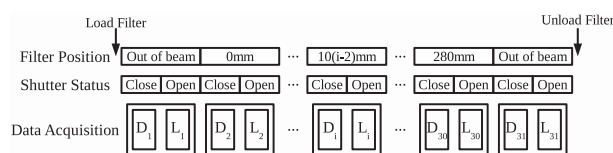


Fig. 6. A sequence of measurements on a radial vector. A filter with filter frame is manually loaded, unloaded, and rotated in the system. Filter movement, shutter control, and data acquisition are done automatically.

stable ($\pm 0.3\%$ per hour). We decided to take the reference data just before and after the transmission measurement.

The standard sequence of transmission measurements is shown in figure 6. Initially, the filter being tested is out of the beam to obtain the reference data pair: Dark (D) and Light (L). Then, the filter is moved into the beam by the motorized stage. We start the filter measurement from its center position. By moving the filter position horizontally in steps of 10 mm, we obtain the pair of D and L data. Twenty-nine data pairs ranging from 0 mm (center of the filter) to 280 mm (near the edge of the filter) with an about 25 mm beam size are sufficient to cover the entire filter aperture, which is about 574 mm in diameter. After the measurements, the reference data are taken after shifting the filter out of the beam.

Each of dark and light data is actually derived from the central 80% clipped mean of 100 spectra taken independently in order to eliminate non-statistical events such as a radiation event on the CCD of the spectrometer. As the exposure time is set at 100 ms for broad-band filters, this procedure takes about 15 min to perform.

After measuring from one position angle of the radial profile, we rotate the filter manually, and it takes about 5 min. We usually measure from eight position angles to cover the entire filter surface. Therefore, it takes about 2.7 hr to obtain data pairs at 232 points (see figure 7) on the surface of a filter with 16 reference data pairs.

In this procedure, we measure at the center of the filter eight times. These eight central data should agree and therefore we can detect the case of an error occurring in the measurement procedure. Figure 8 shows an example of the repeatability. The observed wavelength difference is 0.15 nm at maximum, a tenth of the wavelength resolution of the mini-spectrometer.

3.3 Data reduction

The instrumental profile (IP) of the mini-spectrometer is not sufficiently sharp, especially in the near-infrared region (see figure 9). Therefore, we try to deconvolve the measured spectra by using a Wiener filter (WF) that minimizes the mean square error between the original data and the deconvolved data.

⁵ (<https://www.hamamatsu.com/jp/en/C10083CAH.html>).

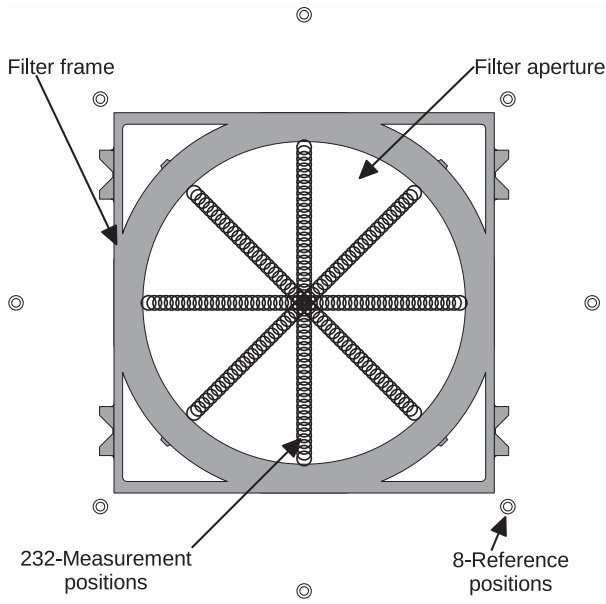


Fig. 7. Two hundred and thirty-two measurement positions on a filter surface with 8 reference positions. The central position is measured 8 times and these data can be used to check the soundness of measurements.

To obtain the IP, which is required as a point spread function (PSF) by the WF, we use a Shimadzu UV3100-PC monochrometer in the Optical shop of the ATC at NAOJ as a source of wavelength-tunable monochromatic light. We set the bandwidth of the monochrometer to 0.5 nm and measure the sample beam of the monochrometer with the mini-spectrometer. We obtained IP data for every 5 nm in 380–1045 nm, and use an appropriate IP as a PSF in the WF. The deconvolved spectrum S_i is represented by the following equation:

$$S_i = \mathcal{F}^{-1} \left[\left(\frac{\mathcal{F}[p]^*}{|\mathcal{F}[p]|^2 + \Gamma} \right) \mathcal{F}[L_i - D_i] \right] (1 + \Gamma), \quad (1)$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the Fourier and inverse Fourier transforms, the superscript $*$ denotes complex conjugate, p is the IP, i represents the number of the data pair, and L_i and D_i are the light and dark data for data pair number i . The index i in this equation is incremented by 1 from 1 to 31 as shown in figure 6. The noise term Γ in the WF was set to such a value that the out-of-band noise level of the deconvolved spectrum is similar to that of the original one. By using these S_i , the filter transmittance can be represented as the following equation:

$$T_i = \frac{S_i}{0.5(S_1 + S_{31})}, \quad (2)$$

where T_i represents the filter transmittance curve data at data number i . S_1 and S_{31} are the continuum data before and after the filter measurement. Though the continuum source

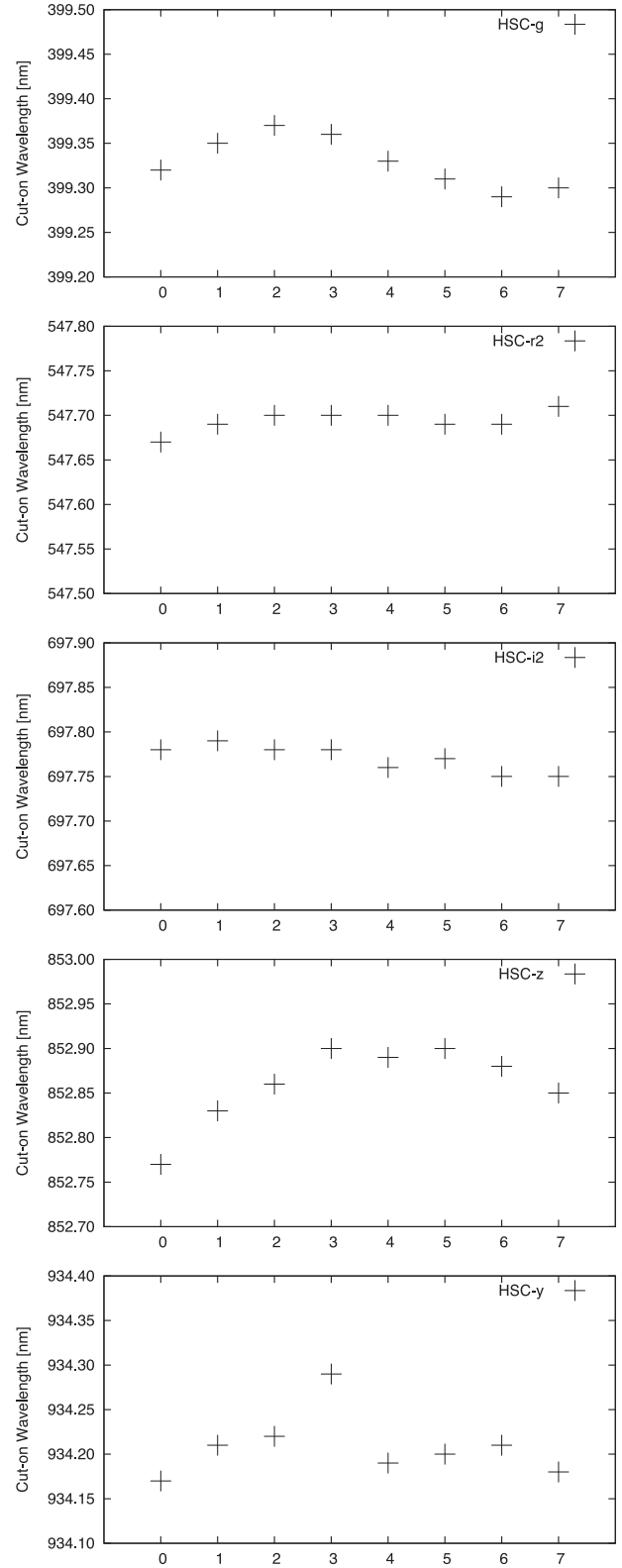


Fig. 8. Eight measurements of cut-on wavelength for each filter.

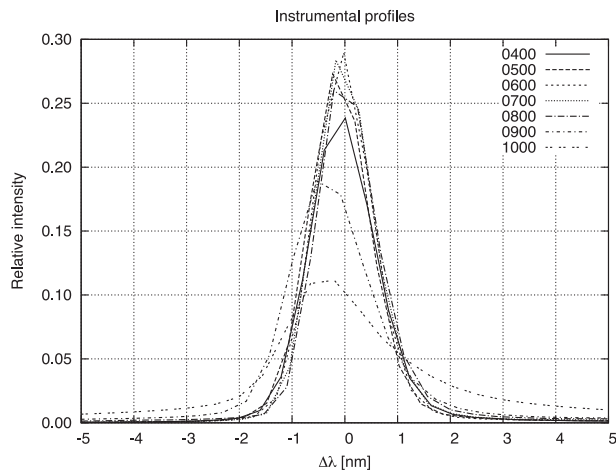


Fig. 9. Examples of the instrumental profile.

is stable, this averaged continuum value data, $(S_1 + S_{31})/2$, is more reliable as a continuum. Therefore, the index i is incremented by 1 from 2 to 30 in equation (2). In our measurement sequence, $r = (i - 2) \times 10$ represents the radius of the data measurement point in millimeters. We calculated the peak transmittance (T_p), cut-on wavelength (λ_{on}), and cut-off wavelength (λ_{off}) from the transmittance curve; that is, λ_{on} and λ_{off} are defined by these two equations: $T(\lambda_{on}) = T(\lambda_{off}) = 0.5 \times T_p$ and $\lambda_{on} < \lambda_{off}$. The central wavelength (CW) and the band width (BW) are derived by $0.5(\lambda_{on} + \lambda_{off})$ and $\lambda_{off} - \lambda_{on}$, respectively.

3.4 Limitations of the system

There are some levels of scattered light in a light spectrum that is formed in the mini-spectrometer. Therefore, we cannot define the out-of-band light leak properties (optical densities, OD) of a filter by this system. We regard that the shipment inspection data from the filter vendors are correct; however, the light leak data should be double-checked. To check the light leak data, we use a Shimadzu SolidSpec3700 monochromator in the Optical shop of the ATC at NAOJ, which has a sufficiently large sample room to load the filter. Though the sample position is limited to the vicinity of the center of the filter and the wavelength coverage is limited to 300–800 nm, the SolidSpec 3700 can find light leaks at the level of OD4 (0.01%), and we cannot find any out-of-band leakage in our broad-band filters. Available leakage data are shown in figure 10.

Since the wavelength range of the mini-spectrometer is limited to about 1080 nm, the shorter, red edge of the HSC-y filter is not covered completely. However, the CCDs of HSC have very little sensitivity at these wavelengths (Kamata et al. 2014) and the transmittance uncertainty of

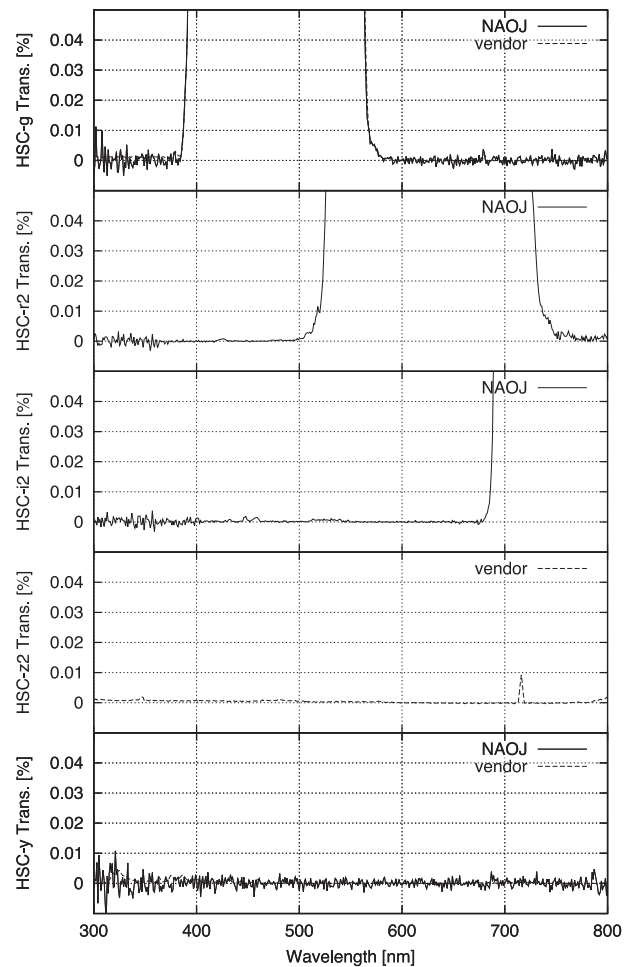


Fig. 10. Available light leak data for the five current broad-band filters. The specifications are $<0.1\%$ for 300–350 nm and $<0.01\%$ for 350–1100 nm. There is no excess of out-of-band transmission.

the red edge of the HSC-y filter has a very small impact on the total system response.

The beam that passes through the filter is the same at each radial position and it cannot simulate the non-telecentricity of telescope optics. We calculated the effect of non-telecentricity on the wavelength shift of a filter. We found that the shift of the central wavelength of HSC-r between the center and the vicinity of the edge of a filter is 0.25 nm, assuming that the effective refraction index of coatings is 2.0. That is still smaller than the expected non-uniformity of coatings; accordingly, we did not take this effect into account.

4 Commissioning data and replacement of two filters

4.1 Filter naming convention

We defined the names of the broad-band filters as HSC-g, HSC-r, HSC-i, HSC-z, and HSC-y. The prefix “HSC-” is mandatory to avoid confusion with similar filters installed

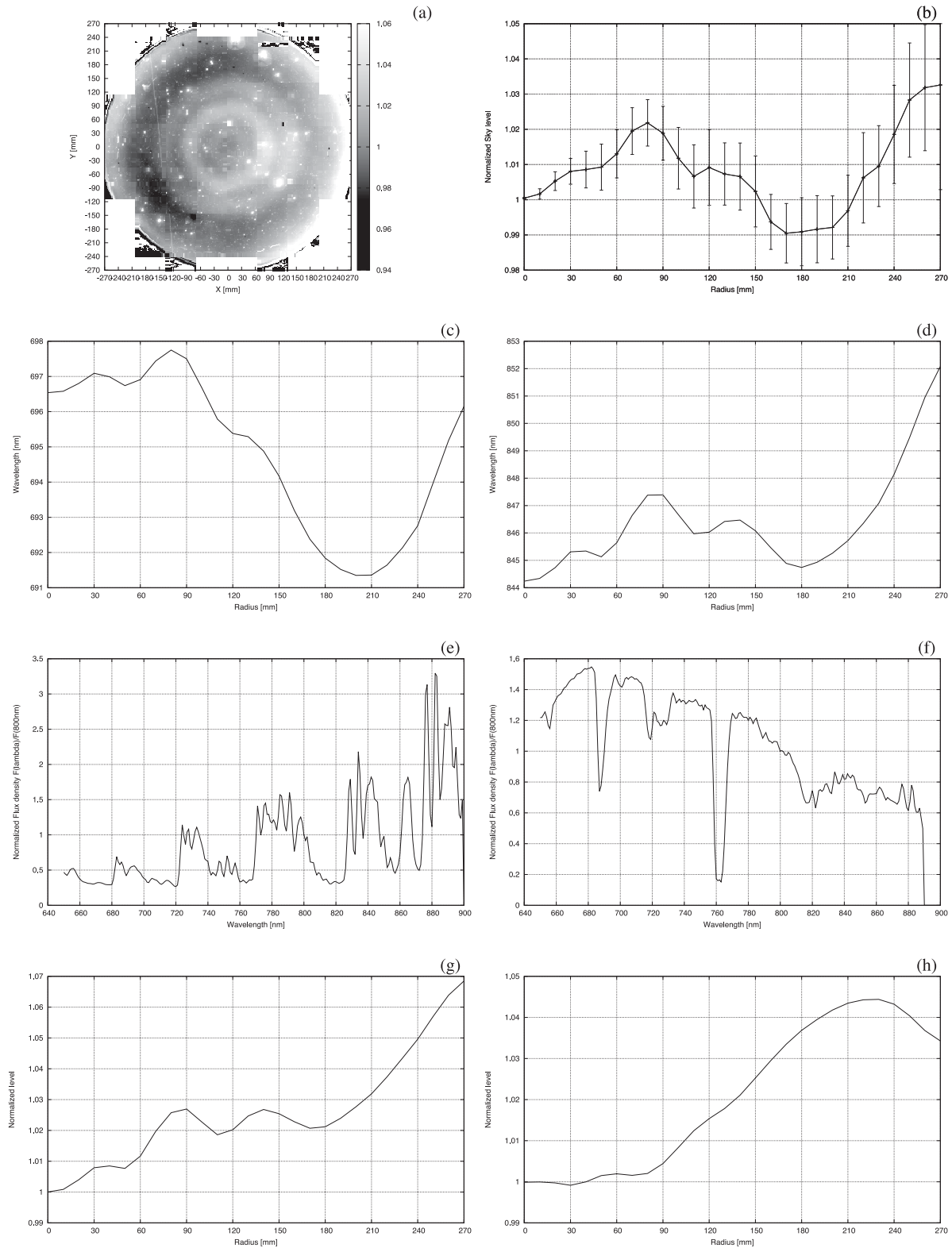


Fig. 11. Figures related to the flat-field error. (a) Representative residual patterns seen on the science exposure of HSC after being divided by the twilight flat. (b) Radial profile of ratio of sky to twilight in the *HSC-i* band. To define a data point at r_i , data in a ring area $r_i - \frac{\Delta r}{2} \leq r < r_i + \frac{\Delta r}{2}$ are processed. The sigma-rejection method is used iteratively to remove outliers. The data points (+) and error bars in the figure show the circular-averaged value and standard deviation, respectively. (c) Rotationally averaged cut-on wavelength as a function of the radial position of the *HSC-i* filter. (d) Rotationally averaged cut-off wavelength as a function of the radial position of the *HSC-i* filter. (e) Spectrum of night sky in the *HSC-i* band. Data from Patat, Ugolnikov, and Postlyakov (2006). (f) Spectrum of twilight in the *HSC-i* band. Data from Patat, Ugolnikov, and Postlyakov (2006). (g) Simulated radial profile of night-sky observation in the *HSC-i* band. (h) Simulated radial profile of twilight observation in the *HSC-i* band.

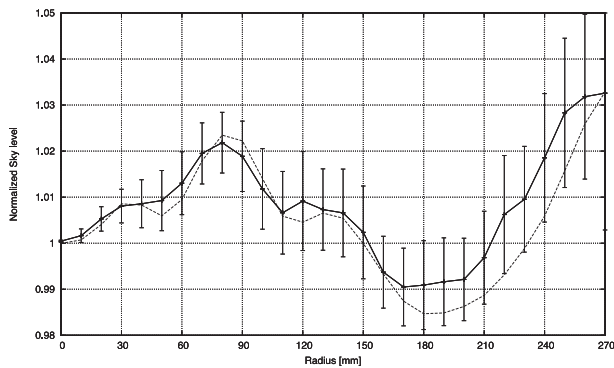


Fig. 12. Radial profile of the ratio of sky to twilight in the *HSC-i* band (+ symbol and solid line) with a simulated profile (dashed line).

in other instruments. The narrow-band filter, whose central wavelength is set at the rest-frame $H\alpha$ emission wavelength, is named *NB0656*. Also, as described below, two broad-band filters named *HSC-r2* and *HSC-i2* have been introduced so as to replace the *HSC-r* and *HSC-i* filters.

4.2 Flat-fielding error caused by the non-uniformity response of filters

Soon after the commissioning run started in 2013, we noted that the science images in the *HSC-i* band divided by a flat image generated from twilight sky images showed concentric residuals, as shown in figures 11a and 11b. This residual pattern can be seen even if we adopt the dome flat instead of the twilight flat.

We originally thought that this was caused by stray light and tried to identify the possible light sources, without success. It turned out that the radial non-uniformity in the filter response caused the residual structure as explained here. The cut-on and cut-off wavelengths of the transmission curve have an axisymmetric spatial variation whose radial profile is shown in figures 11c and 11d.

Meanwhile, the spectrum of the night sky (figure 11e) has stronger band features than the twilight sky (figure 11f). Figures 11g and 11h demonstrate that the non-uniformity coupled with the difference between the two kinds of spectra indeed explains the concentric residual background on the flat-fielded night-sky images. These profiles are calculated by integrating a product of the night-sky or twilight spectrum multiplied by a filter function over the range of the filter transmission. We normalized this integrated value to unity at the center of the filter. The sampling rate of the night-sky and twilight spectrum and the filter transmission was 1 nm. In figure 12, simulated and observed radial profiles of the ratio of sky to twilight in the *HSC-i* filter are shown. The simulated one reproduces the observed profile very well.

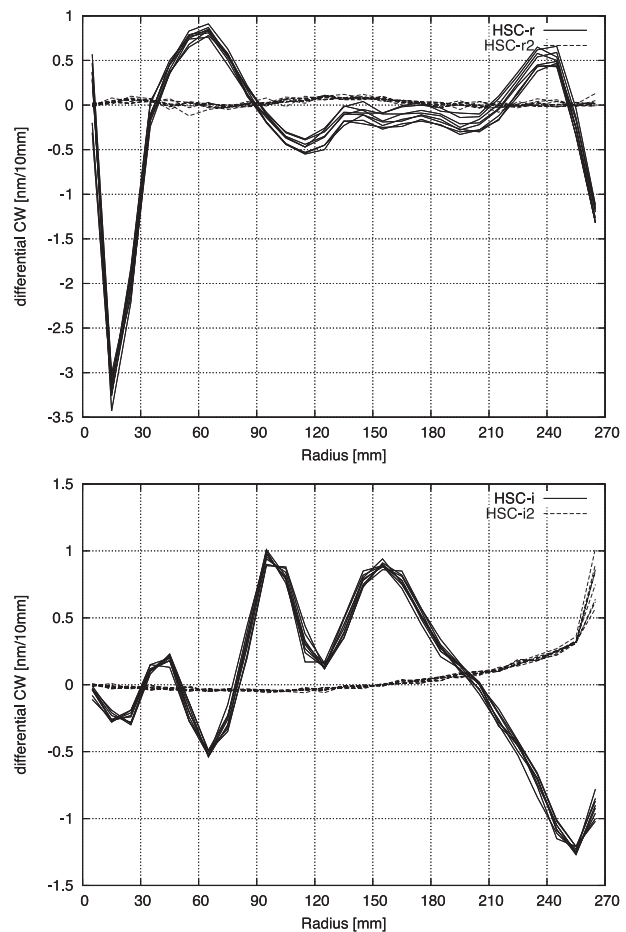


Fig. 13. Differential CW variations as a function of radius. Upper panel: *HSC-r*/*HSC-r2*. Lower panel: *HSC-i*/*HSC-i2*.

4.3 Replacement of two filters

We noted that filters that were delivered in the later phase showed better performance in the positional uniformity of the transmission, reflecting perhaps the maturity of the vendors' deposition technology. We therefore decided to rebuild the *HSC-i* and *HSC-r* filters that had been built in the early phase.

The replacements for these filters, *HSC-r2* and *HSC-i2*, with improved uniformity in the transmission curve, were produced in the fiscal year of 2015. The *HSC-i2* filter was installed into HSC on 2015 November 4 for testing. The observations after 2016 February 2 have been done using *HSC-i2* instead of *HSC-i*. *HSC-r2* was installed on 2016 June 24 and has replaced the textitHSC-r filter since then.

In figure 13 we show the differential variation of the CW for *HSC-r*, *HSC-r2*, *HSC-i*, and *HSC-i2*. The short-scale variation of the CW is much smaller for the two new filters.

When we measured the filter responses for their acceptance with the old measurement system that did not use

Table 4. Properties of HSC filters for five HSC broad-band filters and a narrow-band filter *NB0656* at 656 nm.*

Filter	Production year	Process	Measurement dates	CW [nm]	BW [nm]	T_p [%]	$R(CW)$ [nm]	$R(BW)$ [nm]	$R(T_p)$ [%]
<i>HSC-g</i>	2012	MS+PR	2017 May 31, July 14	472.6	146.8	98.1	1.5	1.1	1.1
<i>HSC-r2</i>	2015	MS+PR	2016 October 27	622.6	150.8	98.4	1.4	2.4	0.4
<i>HSC-i2</i>	2015	MS	2016 November 15	776.7	157.9	94.3	1.6	3.6	2.3
<i>HSC-z</i>	2012	MS+PR	2016 October 26	892.1	79.3	99.4	1.3	3.0	0.6
<i>HSC-y</i>	2012	MS+PR	2016 October 27	1005.1	141.4	97.2	3.4	9.9	1.7
<i>NB0656</i>	2013	IAD	2017 April 26	656.3	11.0	96.4	3.5	0.3	5.3
<i>HSC-r</i>	2010	IAD	2017 March 30	621.5	150.8	90.6	5.1	8.5	6.4
<i>HSC-i</i>	2010	MS	2017 January 05	769.5	153.0	96.8	6.2	8.2	0.9

**HSC-g*, *HSC-r2*, *HSC-i2*, *HSC-z*, *HSC-y*, and *NB0656* are the filters in use. Properties for the obsolete *HSC-r* and *HSC-i* are also presented. Columns CW, BW, and T_p show area-weighted properties. $R(X)$ represents the range of X , $\max(X) - \min(X)$, over the measured area.

an imaging spectrometer but a monochrometer to scan the wavelength, we selected very coarse spatial sampling to save time. However, it was too coarse to find the features such as the radial variation seen in figures 11c and 11d.

5 Properties of HSC filters

Here we summarize the properties of the HSC filters. The names of the filters, production years, processes that were used in production, dates of measurements, CW, BW, T_p , and range of CW, BW, T_p over the measured area are shown in table 4. The *HSC-g*, *HSC-r2*, *HSC-z*, and *HSC-y* filters were made by a process of magnetron sputtering with planetary rotation (MS + PR). The *HSC-i2* and *HSC-i* filters were produced by magnetron sputtering with on-axis rotation (MS). Ion-beam-assisted deposition with on-axis rotation (IAD) was used to make the *NB0656* and *HSC-r* filters. Each of the three processes has its own characteristics. In the filters produced by IAD, short-scale variations of CW can be seen. Magnetron sputtering can create very good axisymmetric coatings; however, some gradual change can be seen in the radial profile of CW, BW, and T_p . The MS+PR process makes a relatively flat profile of characteristics, but it is slightly inferior to MS in asymmetry.

5.1 Area-weighted mean transmittance

We show a summary figure of the filter transmission curves in figure 14. In the measurement of the transmission, it is important to minimize the slope of the continuum of the light source in the pass-band. Otherwise, we would have a wavelength-dependent systematic error in transmittance. We used the deuterium lamp for the *HSC-g* filter because of a lack of blue light in the tungsten-halogen lamp, while the tungsten-halogen lamp was used for the other filters. Because the deuterium lamp is somewhat dimmer than the

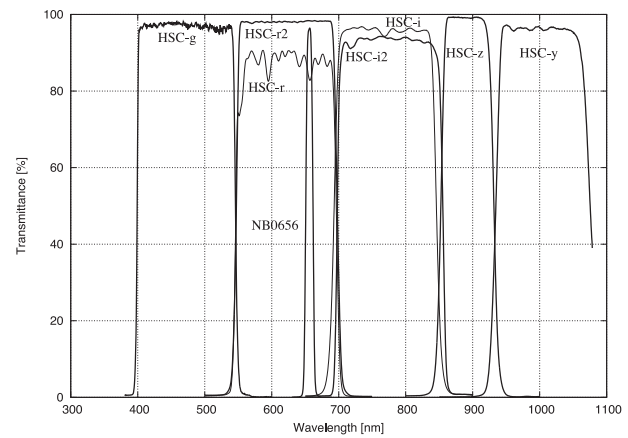


Fig. 14. Realized transmission curves of the HSC filter set. Area-averaged transmission data of each filter are plotted. The red side of *HSC-y* is cut off because of the limit of the measurement system.

tungsten-halogen lamp, certain noise levels can be seen in the measurement of the *HSC-g* pass-band; nevertheless, the cut-on and cut-off wavelengths are defined very well. The *HSC-r2* filter shows an improvement in transmittance over *HSC-r*. The *HSC-i2* is slightly inferior to *HSC-i* in transmittance; however, the decrease in transmittance is negligible compared with the improved uniformity in the transmittance curve across the field of view (see figure 15). The *HSC-z* has very good transmittance, and the transmittance of *HSC-y* is acceptable.

5.2 Surface uniformity

The range (R) of T_p for *HSC-g* seems to be affected by its noise (see figure 15). If we carry out a better signal-to-noise ratio measurement, it is probable that we will get a smaller $R(T_p)$ value. The *HSC-r2* and *HSC-i2* filters show good improvement in their surface uniformity of CW and BW.

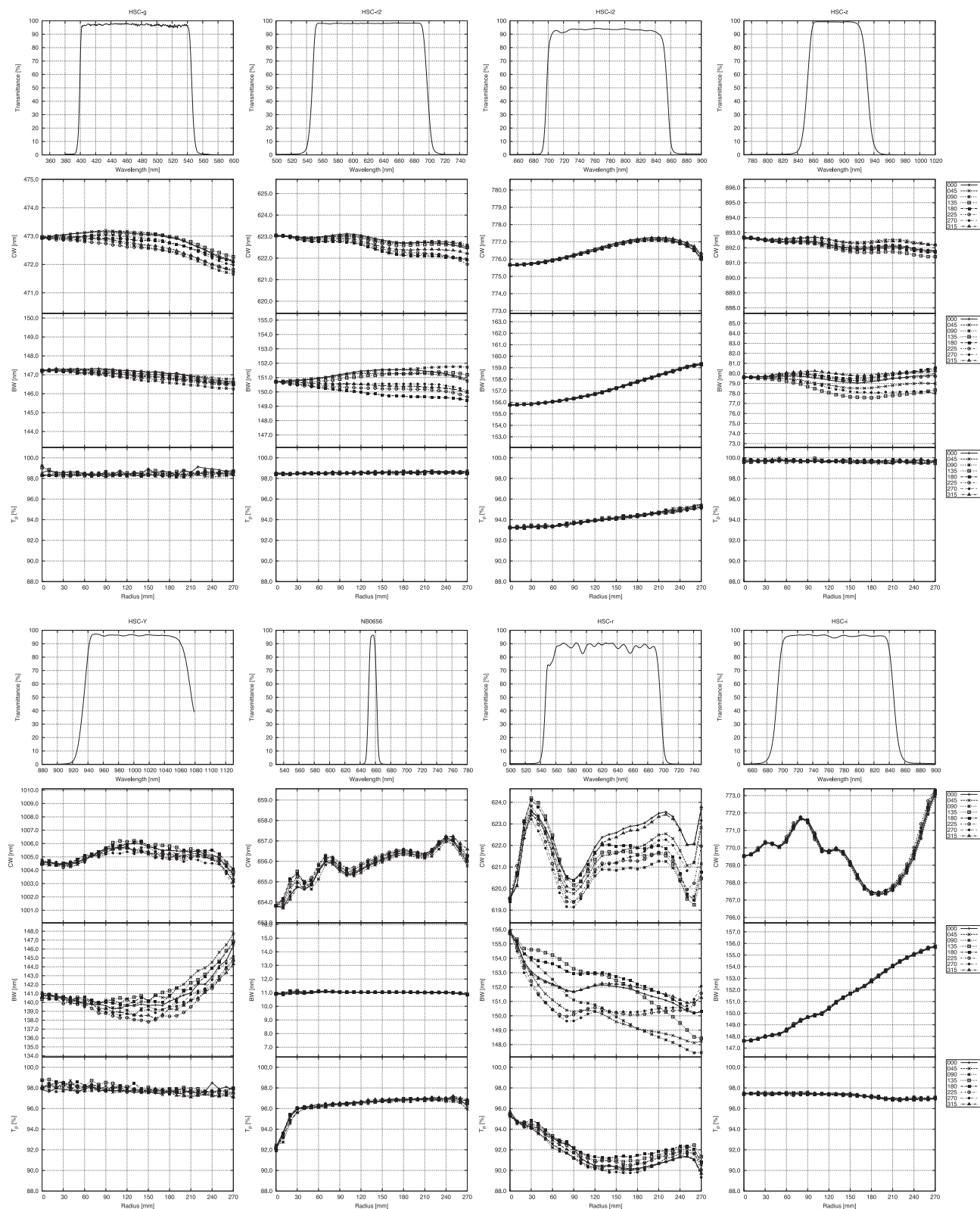


Fig. 15. Area-weighted mean of transmittance curve and radial profiles of CW, BW, and T_p for the HSC filters.

Table 5. Area-weighted mean transmittance data for seven broad-band filters.

λ	HSC-g	HSC-r2	HSC-i2	HSC-z	HSC-y	HSC-r	HSC-i
380	0.65						
385	0.58						
390	0.65						
395	3.24						
400	71.97						
405	96.64						
410	97.88						
415	96.94						
420	97.19						
425	97.30						
430	96.79						
435	97.78						
440	97.34						
445	97.62						
450	98.17						
455	97.67						
460	97.49						
465	97.43						
470	97.04						
475	96.50						
480	97.98						
485	97.07						
490	97.24						
495	96.72						
500	96.44						
505	96.59						
510	96.41	0.44				0.58	
515	96.11	0.48				0.57	
520	95.50	0.53				0.57	
525	96.07	0.62				0.59	
530	96.63	0.79				0.66	
535	96.16	1.66				1.00	
540	97.28	5.30				3.57	
545	63.51	28.11				31.50	
550	9.66	82.97				74.31	
555	1.44	98.12				76.44	
560	0.85	98.00				86.53	
565	0.57	97.95				88.62	
570	0.19	97.62				90.45	
575	0.15	98.03				88.68	
580		97.97				86.81	
585		98.19				90.02	
590		97.83				88.55	
595		97.98				82.49	
600		97.98				88.19	
605		98.01				89.91	
610		98.10				87.82	
615		97.99				90.41	
620		98.20				88.99	
625		98.11				90.53	
630		97.95				89.97	
635		98.08				90.02	
640		98.03				86.65	
645		98.12				87.98	
650		98.28				88.91	
655		98.40				83.79	
660		98.27				84.69	
665		98.35				88.64	
670		98.20	0.35			86.99	0.72
675		98.20	0.37			89.50	1.26
680		98.31	0.40			87.61	3.18
685		98.31	0.41			87.11	9.52
690		96.92	1.00			86.07	28.15
695		74.79	15.59			62.85	63.07
700		31.95	74.30			18.83	88.83
705		5.86	90.18			2.29	94.94

Table 5. (Continued)

λ	HSC-g	HSC-r2	HSC-i2	HSC-z	HSC-y	HSC-r	HSC-i
710		1.69	92.76			0.57	95.85
715		0.72	91.59			0.31	96.12
720		0.37	91.36			0.26	96.44
725		0.31	92.45			0.23	96.47
730		0.28	93.75			0.22	96.51
735			93.76				96.81
740			93.66				96.25
745			93.55				95.93
750			93.38				96.24
755			93.76				96.58
760			94.23				95.85
765			94.25				94.64
770			93.93				94.35
775			93.78				95.39
780			93.68				96.25
785			93.43				95.81
790			93.51				95.39
795			93.89				95.42
800			94.06				95.85
805			93.64				96.41
810			93.21	0.42			96.20
815			93.28	0.44			95.66
820			93.24	0.45			95.63
825			92.74	0.53			95.79
830			92.59	0.72			95.85
835			92.72	1.12			94.83
840			91.82	2.63			85.61
845			89.84	9.58			55.99
850			84.02	31.67			23.96
855			53.68	70.10			8.10
860			10.51	96.51			2.97
865			1.92	99.21			1.45
870			0.95	99.27			0.95
875			0.78	99.24			0.83
880			0.79	99.22			0.81
885			0.76	99.11			0.76
890			0.80	99.09			0.72
895				99.09			
900				99.10	0.18		
905				99.35	0.28		
910				99.11	0.43		
915				98.65	0.77		
920				96.63	2.22		
925				87.93	8.62		
930				62.71	25.74		
935				26.44	52.20		
940				7.91	82.22		
945				2.39	96.41		
950				0.73	97.22		
955				0.33	96.83		
960				0.17	95.80		
965				0.13	96.19		
970				0.13	96.44		
975					96.41		
980					96.42		
985					95.76		
990					96.12		
995					96.71		
1000					96.54		
1005					95.80		
1010					95.96		
1015					96.59		
1020					96.77		
1025					96.18		
1030					96.13		
1035					96.37		
1040					96.03		

Table 5. (Continued)

λ	<i>HSC-g</i>	<i>HSC-r2</i>	<i>HSC-i2</i>	<i>HSC-z</i>	<i>HSC-y</i>	<i>HSC-r</i>	<i>HSC-i</i>
1045					95.85		
1050					95.16		
1055					94.35		
1060					91.49		
1065					84.60		
1070					70.99		
1075					51.42		

In particular, short-scale variations in the radial profile are well suppressed in these filters (see figure 15). The *HSC-z* also seems good for surface uniformity. Slightly large variations of CW and BW in *HSC-y* are caused by difficulties in coating at the long-wavelength range near 1000 nm, since the total thickness of the coating increases in proportion to the wavelength of concern.

6 Conclusion

We developed a standard set of HSC filters which comprised five broad-band and one narrow-band filters. These filters are produced as purely interference filters. At first, standard evaporation coating did not satisfy our requirement, but finally we obtained sufficiently good filters by employing sputtering, IAD, and ion plating. To measure the transmittance properties of HSC more efficiently, we built a semi-automatic transmission measurement system and placed it in the summit facility of the Subaru Telescope. We have measured five broad-band filters which are active in observations, two obsoleted broad-band filters, and a narrow-band filter, and we find that the five active broad-band filters and the narrow-band filter meet their specifications. The system enables monitoring of the transmittance of filters and assists us in securing the soundness of survey observations.

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Table 6. Area-weighted mean transmittance data for the narrow-band filter *NB0656*.

λ	<i>NB0656</i>	λ	<i>NB0656</i>
635	0.09	656	96.28
636	0.09	657	96.50
637	0.09	658	95.38
638	0.10	659	91.53
639	0.11	660	80.45
640	0.12	661	64.07
641	0.15	662	43.31
642	0.18	663	23.08
643	0.22	664	9.96
644	0.32	665	3.70
645	0.56	666	1.76
646	0.95	667	0.98
647	2.55	668	0.60
648	6.95	669	0.40
649	15.86	670	0.28
650	32.77	671	0.20
651	53.17	672	0.15
652	74.45	673	0.11
653	88.56	674	0.08
654	94.39	675	0.05
655	95.75		

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Appendix. Area-weighted transmittance data

The area-weighted mean transmittance curve and CW, BW, and T_p of the broad-band filters and the narrow-band filter for the HSC are shown in figure 15. Numerical data for these filters are shown in tables 5 and 6. Data points for the broad-band filters and the narrow-band filter *NB0656* are listed for every 5 nm and 1 nm, respectively.

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